

Functional Annotation Report: *carB* (Q88DU6) — Carbamoyl Phosphate Synthetase Large Chain, *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

carB (PP_4723; UniProt Q88DU6) encodes the large (catalytic) subunit of carbamoyl phosphate synthetase (CPS; EC 6.3.4.16 / 6.3.5.5). Together with the glutaminase small subunit (*carA*), it catalyzes the ATP-dependent synthesis of **carbamoyl phosphate** in the cytoplasm from bicarbonate, two molecules of MgATP, and ammonia derived from glutamine:



Carbamoyl phosphate is the single shared precursor that feeds **both the arginine biosynthetic pathway and the pyrimidine nucleotide biosynthetic pathway**. The large subunit itself performs the two ATP-dependent phosphorylation steps (bicarbonate → carboxyphosphate; carbamate → carbamoyl phosphate) at two internal ATP-grasp active sites, and carries the C-terminal allosteric domain through which the enzyme is feedback-regulated.

Gene identity is confirmed: the symbol *carB*, the organism (*P. putida* KT2440), the CarB family / HAMAP rule MF_01210, and the ATP-grasp + CPSase domain architecture all match the UniProt target. No gene-symbol ambiguity was encountered — *carB* is a universally conserved, well-characterized enzyme name.

2. Protein Identity and Domain Architecture (direct analysis of Q88DU6)

Sequence analysis of Q88DU6 (retrieved from UniProt) confirms a **1,073-residue, ~117 kDa** protein with the canonical CPS large-subunit organization:

Region	Residues	Role
Carboxyphosphate synthetic domain (ATP-grasp 1: 133–328)	1–403	Phosphorylation of bicarbonate → carboxyphosphate; ammonia attack → carbamate
Oligomerization domain	404–553	Subunit/oligomer contacts
Carbamoyl phosphate synthetic domain (ATP-grasp 2: 678–869)	554–935	Phosphorylation of carbamate → carbamoyl phosphate
Allosteric / MGS-like domain	936–1073	Binds allosteric effectors (UMP, IMP, ornithine)

Domain signatures: ATP-grasp (IPR011761), CPSase large-subunit (IPR006275), pre-ATP-grasp (IPR058047), and MGS-like (IPR011607; Pfam PF02142); governed by HAMAP rule **MF_01210 (CarB family)**. Numerous nucleotide/substrate binding-site residues are annotated within **both** ATP-grasp folds, confirming two independent catalytic sites. This architecture is a one-to-one match to the crystallographically characterized *E. coli* enzyme, so its detailed mechanism transfers with high confidence.

Ortholog-level confirmation (this work): A global (Needleman–Wunsch) alignment of Q88DU6 against *E. coli* CarB (P00968) shows **78.9 % sequence identity over the full length** (both proteins are exactly 1,073 residues). Experimentally defined *E. coli* functional residues are conserved in *P. putida*: the UMP/IMP effector anchor **Lys993→Lys992**, IMP anchors **Lys954→Lys953** and **Asn1015→Asn1014**, and the ornithine-activator/signal-transmission residues **Glu783→Glu782**, **Glu892→Glu891**, **Thr1042→Thr1041**, plus K-loop **Glu761→Glu760**. The one notable exception is the IMP photoaffinity-labeling residue **His995, which is a threonine (His995→Thr994)** in *P. putida* — predicting a possibly attenuated/altered IMP activation while UMP inhibition and ornithine activation are retained. This near-identity (same length, ~79 % identity, conserved active-site chemistry) makes transfer of the *E. coli* mechanism to *P. putida* essentially certain.

Catalytic-site conservation (this work): Mapping the *E. coli* catalytic/Mg²⁺-coordinating residues onto Q88DU6 shows **9/9 probed active-site residues are identical** — N-terminal carboxyphosphate site (Glu215, Glu299, Asn301, Glu383) and C-terminal carbamoyl-phosphate site (Glu760/K-loop, Glu840, Asn842, Glu915) — demonstrating that **both** ATP-dependent phosphorylation active sites are structurally and catalytically intact in *P. putida*. This upgrades the mechanistic transfer from "family-level inference" to "conserved-active-site evidence."

Curated pathway/quaternary annotations (UniProt): Both catalytic activities are annotated — glutamine-dependent ($\text{HCO}_3^- + \text{L-Gln} + 2 \text{ ATP} \rightarrow \text{carbamoyl phosphate} + \text{L-Glu} + 2 \text{ ADP} + \text{P}_i$) and ammonia-dependent ($\text{HCO}_3^- + \text{NH}_4^+ + 2 \text{ ATP} \rightarrow \text{carbamoyl phosphate} + 2 \text{ ADP} + \text{P}_i$). Pathway assignments: **L-arginine biosynthesis — carbamoyl phosphate from bicarbonate, step 1/1;** and **pyrimidine (UMP) de novo biosynthesis — step 1/3** toward (S)-dihydroorotate. The holoenzyme is a **tetramer of heterodimers, ($\alpha\beta$)₄** (KEGG ppv:PP_4723; eggNOG COG0458).

3. Primary Function — the Reaction and Its Chemistry

CPS assembles carbamoyl phosphate through a **minimum of four consecutive reactions involving three unstable intermediates** (ammonia, carboxyphosphate, carbamate) [PMID 9818189, 10387030]:

1. **Glutamine hydrolysis** (small subunit, carA): glutamine $\rightarrow$ glutamate + NH_3 via a Cys269 thioester intermediate. (This is the small subunit's job; the ammonia is channeled to the large subunit.)
2. **Bicarbonate phosphorylation** (large subunit, N-terminal ATP-grasp): $\text{ATP} + \text{HCO}_3^- \rightarrow \text{carboxyphosphate} + \text{ADP}$ [PMID 9818189].
3. **Carbamate formation**: NH_3 attacks carboxyphosphate $\rightarrow$ carbamate + P_i [PMID 10387030].
4. **Carbamate phosphorylation** (large subunit, C-terminal ATP-grasp): $\text{ATP} + \text{carbamate} \rightarrow \text{carbamoyl phosphate} + \text{ADP}$ [PMID 9818189].

Substrate specificity / nitrogen source: CPS can use either glutamine (physiological, EC 6.3.5.5) or free ammonia (EC 6.3.4.16) as the nitrogen donor; the large subunit strictly requires ammonia, which in vivo is supplied by the carA glutaminase [PMID 11212301, 10387030]. Bicarbonate (not CO_2) is the carbon/oxygen source, and two MgATP are consumed per product. K^+ is a required activating monovalent cation.

Evolutionary/structural inference: The large subunit is built as two homologous halves related by a near-exact internal 2-fold axis, indicating evolution from a homodimeric ancestor; each half's first three domains resemble biotin carboxylase (the ATP-grasp fold) [PMID 9174345].

4. Where the Function Is Carried Out (localization and channeling)

CPS is a **soluble cytoplasmic** biosynthetic enzyme. A defining structural feature is that its **three active sites are widely separated (~45 Å apart) and connected by an internal molecular tunnel spanning ~96–100 Å**; reactive intermediates travel this tunnel rather than being released to solvent [PMID 9174345, 11212301, 9818189]. This intramolecular channeling protects the chemically labile carboxyphosphate and carbamate. Catalysis at the three sites is kinetically synchronized: glutamine hydrolysis is stimulated ~275-fold by ATP/bicarbonate turnover, coordinating the small-subunit and large-subunit chemistry [PMID 10819970].

5. Pathway Context and Regulation

Carbamoyl phosphate stands at the **crossroad of arginine and pyrimidine biosynthesis**; in *E. coli* and most other Gram-negative bacteria (including *P. putida*) it is produced by a single CPSase encoded by the **carAB operon** [PMID 30238253]. The *P. putida* KT2440 *carB* (PP_4723) is the large-subunit gene of this operon.

Operon / partner subunit (this work): In the KT2440 genome, *carB* (PP_4723) is immediately adjacent to **PP_4724 = *carA* (Q88DU5)**, the glutaminase small chain (378 aa, EC 6.3.5.5), confirming a single, co-localized *carA/carB* gene pair that encodes one glutamine-dependent CPS — the Gram-negative paradigm, in contrast to *Bacillus* (two pathway-specific isozymes) [PMID 12561954]. In vivo, CarB's ammonia substrate is supplied by CarA's hydrolysis of glutamine; free NH_4^+ can substitute in vitro (accounting for the dual EC 6.3.5.5 / 6.3.4.16 assignment). (The upstream gene PP_4722 is the unrelated *greA*.)

Because one metabolite serves two pathways, enzyme activity is **allosterically tuned by effectors from both consuming pathways plus purine metabolism**, all acting through the C-terminal allosteric domain (conserved as res 936–1073 in Q88DU6): - **UMP** (pyrimidine end-product) — **inhibitor** - **Ornithine** (arginine pathway intermediate) — **activator** - **IMP** (purine) — **activator**

Structural studies localize IMP binding to the C-terminal β -sheet domain (Ser937–Lys1073), with Lys993/His995 crosslinked by UMP/IMP in overlapping sites [PMID 10428826, 10194302].

Pseudomonas-specific regulation: In *Pseudomonas*, the *carAB* operon (containing *carB*) is transcribed as a single unit **controlled by both arginine and pyrimidines** — arginine modulates transcription initiation while pyrimidine control is exerted post-initiation [PMID

8169201]. *carA* and *carB* are contiguous in *P. stutzeri* and in *P. putida* KT2440 (PP_4724/PP_4723), whereas in *P. aeruginosa* they are separated by a 216-aa ORF yet co-transcribed [PMID 8169201]. Notably, an early direct study in *P. putida* found only modest (≤ 5 -fold) repression/derepression of arginine and pyrimidine biosynthetic enzymes [PMID 176312], implying that flux through carbamoyl phosphate is governed substantially by **allosteric modulation of CPS activity** in addition to this relatively loose transcriptional control. Effector binding also modulates oligomeric state: ornithine and IMP favor an $(\alpha\beta)_4$ tetramer, whereas UMP favors an $(\alpha\beta)_2$ dimer [PMID 11551199]. Mutagenesis has mapped the ornithine signal-transmission conduit (E783, E892, T1042, K-loop residues) linking the effector site to the carbamate-phosphorylation active site [PMID 11913967, 11943174].

6. Supported vs. Refuted Hypotheses

Supported: - H1: *carB* is the CPS large (synthetase) subunit catalyzing carbamoyl phosphate formation — **supported** (annotation + orthology + domain match). - H2: The large subunit performs two ATP-dependent phosphorylations at two ATP-grasp sites — **supported** (structure + kinetics; confirmed in Q88DU6 sequence). - H3: Product feeds arginine + pyrimidine pathways; enzyme is allosterically regulated (UMP/ornithine/IMP) via a conserved C-terminal domain — **supported**. - H4: Function is cytoplasmic with internal substrate channeling — **supported**.

Refuted / not applicable: - Gene-symbol ambiguity concern — **refuted**; *carB* unambiguously denotes CPS large chain across bacteria.

7. Limitations and Future Directions

- Nearly all mechanistic and structural evidence derives from the **orthologous *E. coli* enzyme**; no *P. putida*-specific enzymology or crystal structure was located. Transfer of function is justified by very high domain/family conservation (HAMAP MF_01210, one-to-one architecture match) but has not been experimentally verified for KT2440.
- The precise allosteric response profile (effector affinities, magnitude) may differ quantitatively in *P. putida* and warrants direct kinetic characterization.
- Future work: recombinant expression + activity/regulation assays of PP_4723, and structural modeling (e.g., AlphaFold) to confirm the tunnel and effector-site residues in the *P. putida* enzyme.

Key References

- Thoden et al. 1997, *Biochemistry* — CPS X-ray structure, "journey of 96 Å"

([P 9174345](#))

- Holden, Thoden & Raushel 1999 — CPS review, 100 Å molecular tunnel

([P 11212301](#))

- Raushel et al. 1998 — "crooked path," reaction intermediates

([P 9818189](#))

- Raushel, Thoden & Holden 1999 — amidotransferase family / mechanism

([P 10387030](#))

- Miles & Raushel 2000 — synchronization of reaction centers

([P 10819970](#))

- Thoden et al. 1999 — IMP binding structure

([P 10428826](#))

- Bueso et al. 1999 — IMP/UMP overlapping allosteric site, His995

([P 10194302](#))

- Kim & Raushel 2001 — allosteric control of oligomerization

([P 11551199](#))

- Pierrat/Rochera 2002 — ornithine allosteric conduit

([P 11913967](#))

([P 11943174](#))

- Charlier, Nguyen Le Minh & Roovers 2018 — regulation at arginine/pyrimidine crossroad, carAB operon

([P 30238253](#))

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